

LIANGJIE (JEFFREY) CHEN

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RESEARCH INTERESTS	data-driven control theory, mathematical optimization, algebraic graph theory, scientific and high performance computing, power system modeling and optimization, reinforcement learning		
EDUCATION	University of Toronto		
	Ph.D., Electrical and Computer Engineering		In progress
	M.A.Sc., Electrical and Computer Engineering		2022
	B.A.Sc. (Graduated with Honours), Computer Engineering		2020
	Minors in Artificial Intelligence Engineering, Robotics and Mechatronics Engineering		
RESEARCH EXPERIENCE	Graduate Student Researcher	August 2020 – Present	
	<i>Department of Electrical and Computer Engineering, University of Toronto</i>		
	Advisor: Prof. J. W. Simpson-Porco		
	• Current research project: data-driven disturbance rejection and estimation for unknown linear time-invariant systems • Past research project: development and analysis of a fixed-point algorithm for the AC power flow problem		
	Undegraduate Student Researcher	May 2019 – April 2020	
	<i>Department of Electrical and Computer Engineering, University of Toronto</i>		
	Advisor: Prof. J. E. Tate		
	• Acceleration of Newton-Raphson based power flow computations by training convolutional neural nets to generate viable bus voltage values as initial conditions		
	Undergraduate Research Assistant	May – August 2017	
	<i>Department of Mechanical and Industrial Engineering, University of Toronto</i>		
	Advisor: Dr. A. Melnikov		
	• Comparison of Single Frequency-Thermal Wave Radar imaging and Lock-in Thermography on signal-to-noise ratio across wide frequency range and measurement time		
PAPERS	L. Chen and J. W. Simpson-Porco, “Data-driven output regulation using single-gain tuning regulators,” in <i>2023 IEEE Conference on Decision and Control</i> , (Singapore), pp. 2903–2909, Dec. 2023		
	L. Chen and J. W. Simpson-Porco, “A fixed-point algorithm for the AC power flow problem,” in <i>2023 American Control Conference</i> , (San Diego, CA, USA), pp. 4449–4456, May 2023		
	L. Chen and J. E. Tate, “Hot-Starting the Ac Power Flow with Convolutional Neural Networks,” Apr. 2020. arXiv: 2004.09342 [eess.SY]		
	A. Melnikov, L. Chen, D. R. Venegas, K. Sivagurunathan, Q. Sun, A. Mandelis, and I. R. Rodriguez, “Single frequency thermal wave radar: A next-generation dynamic thermography for quantitative non-destructive imaging over wide modulation frequency ranges,” <i>Review of Scientific Instruments</i> , vol. 89, Apr. 2018		

HONOURS AND AWARDS	Postgraduate Scholarship – Doctoral	2024 – 2026
	<i>Natural Sciences and Engineering Research Council of Canada</i> (\$120,000 total)	
	Ontario Graduate Scholarship	2023 – 2024
	<i>Government of Ontario</i> (Awarded twice for \$30,000 total; declined 2024 award for PGS-D)	
	Hatch Graduate Scholarship for Sustainable Energy Research	2022 – 2023
	<i>Hatch Ltd., Canada</i> (Awarded twice for \$20,000 total)	
	Queen Elizabeth II Graduate Scholarship in Science and Technology	2022
	<i>Government of Ontario</i> (\$15,000 total)	
	H. W. Price Research Fellowship in Electrical Engineering	2021
	<i>Hydro One Ltd., Canada</i> (\$4,300 total)	
	Undergraduate Student Research Award	2019
	<i>Natural Sciences and Engineering Research Council of Canada</i> , (\$5,600 total)	
	Dean’s Honours List	2018 – 2020
	<i>Faculty of Applied Science and Engineering, University of Toronto</i>	
	President’s Entrance Scholarship	2016
	<i>Faculty of Applied Science and Engineering, University of Toronto</i> , (\$2,000 total)	
	<i>Faculty of Arts and Science, University of Toronto</i> (\$2,000 total, declined)	
TECHNICAL SKILLS	Programming Languages: MATLAB, Julia, Python, C++, C, L ^A T _E X, Bash Softwares and APIs: • Mathematical Optimization: JuMP, CVX, YALMIP • Power System Analysis: PowerModels.jl, MATPOWER, PSCAD • Machine Learning: Flux.jl, PyTorch • Simulation: Simulink	
TEACHING EXPERIENCE	Teaching Assistant <i>University of Toronto</i> • ECE1659 Robust and Optimal Control (Winter 2024) • ECE557 Linear Control Theory (Fall 2021-24) • ECE311/ECE356 Introduction to Control Systems (Fall/Winter 2022-24) • ECE216 Signals and Systems (Winter 2022-23) • MAT290 Advanced Engineering Mathematics (Fall 2023, Head TA Fall 2024-25) • MAT188 Linear Algebra (Fall 2020-2022)	
PROFESSIONAL TRAINING	Scientific and High Performance Computing	June 2019
	<i>Compute Ontario Summer School</i>	
PROFESSIONAL AFFILIATIONS	Student Member , <i>Institute for Electrical and Electronics Engineers (IEEE)</i> Student Member , <i>Society for Industrial and Applied Mathematics (SIAM)</i>	

VOLUNTEERING AND OTHER EXPERIENCE	Event Planner , <i>IEEE Control System Society/American Control Conference</i>	2024
	Violinist , <i>UofT Campus Philharmonic Orchestra</i>	2021 – Present
	Assistant Concertmaster , <i>Skule Music Chamber Night</i>	2019
	Fabrication Team Member , <i>Blue Sky Solar Racing</i>	2016 – 2017

Last Updated: September 2, 2025